

Designing AIS Link based on Software Defined Radio for LEO Satellites

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Abstract—The Automatic Identification System (AIS) was developed by the International Maritime Organisation (IMO) in order to improve the safety of navigation. However, AIS devices allow identification of vessels in the vicinity within a limited range (15–20 nautical miles). Therefore, a space-based AIS system is used nowadays as a relay to shore stations with a higher range of detection. In this paper, an AIS radio link design based on software defined radio between vessels in Moroccan territorial sea and a specific ground station in Rabat is proposed. The AIS data is received by satellite via a highly integrated software defined radio (SDR) over very high frequency (VHF) channels and downloaded to the ground station via a S-band transceiver as a data communication subsystem. This paper focuses on the preliminary process of designing an optimum link for the space-based AIS system.

Index Terms—CubeSat, AIS, Totem-SDR, S-band, link budget.

I. INTRODUCTION

Maritime vessel collisions increase the occurrence probability of hazardous and noxious substances (HNS) spills, which are devastating for the ocean's environment and harmful for the marine ecosystems. In order to improve navigational safety and reduce the probability of accidents at sea, the Automatic Identification System (AIS) has been mandated by the IMO since 2004. The AIS ensures the exchange of static and dynamic information between ships and shore station located along the coastline in a limited area.

This limitation can be extended by the use of LEO satellites, especially CubeSats, as an optimal relay alternative between ships and ground stations. Ranging from 1 Unit to 12U, CubeSats increase accessibility to space by reducing cost and development time. A 1U CubeSat standard of nanosatellite has a standard shape of 10cm^3 and a 2 kg weight [1].

Ramp up	Training sequence	Start flag	Data	CRC	End flag	Buffer
8 bits	24 bits	8 bits	168 bits	16 bits	8 bits	24 bits

Fig. 1. AIS message format.

The AIS transmission utilizes a non-return-to-zero inverted (NRZI) waveform encoder, the Gaussian minimum shift keying (GMSK) modulation, and a datarate of 9600 bps. With a 12.5 W output power, the resulting AIS signal is transmitted for class A ships and can fit 2250 AIS messages over one minute using a single AIS channel. The information contained in an AIS message is encoded in 256-bit string based on the High-level Data Link Control (HDLC) format, with 168 bits containing the vessel data, as given in Fig. 1.

To avoid interference and data loss, AIS Class A transponders use a system known as Self Organizing Time Division Multiple Access (SOTDMA) [2], in which multiple transponders automatically know how to negotiate with other AIS units the time slots, as well as how to organize themselves if another transponder tries to use the same time slot.

Previous work in the literature [3] discusses the design of the AIS communication subsystem for micro-satellite, which is more expensive with a long development time and minimal launch opportunities. In addition, a study by [4] was performed in order to examine the effects of distance and angle between satellites to ship for computing the probability of detection. In [5], the authors tested in the laboratory a communication system designed for maritime needs using System on Chip (SoC) based on software defined radio (Lime-SDR), which has not a high Technology Readiness Level (TRL) [6].

The main goal of this paper is the preliminary process of AIS communication link design that meets the mission requirements using commercial components proven in space operational environments with high flight heritage. Those devices must guarantee collecting, storing, and downloading AIS data continuously between the space and ground segments.

This document is organized as follows: Section II contains the design requirements and operation concept. Section III goes through the space-based AIS link design for the selected components. Finally, the link budget simulation results are analyzed in the last section.

II. DESIGN REQUIREMENTS AND OPERATION CONCEPT

A. Design requirements

For AIS purposes, the proposed CubeSat employs a Totem Software Defined Radio [7] as a reception (Rx) module for AIS Data Transmission. On the basis of current European Co-operation for Space Standardization (ECSS) [8], some design requirements are given for the proposed satellite.

1. The CubeSat shall have a 3 unit standard.
2. The orbit altitude shall be around 500km.
3. TOTEM-SDR motherboard shall be connected to VHF antenna using VHF front end.
4. The AIS transmission shall be over regular AIS channels (161.975MHz and 162.025MHz) and AIS Long Range channels (156.775MHz and 156.825MHz) using GMSK modulation and 9.6Kbps data rate.
5. The platform shall include S-band transceiver and Patch antenna for payload data downlink.
6. The required link margin shall exceed 6dB and 3dB in uplink and downlink respectively.

B. Operation concept

Equipped with a Software Defined Radio (SDR) as its main payload, this satellite is intended to be used as a data relay for AIS communication. The concept of operation is to cover the territorial seas of Morocco. A preliminary analysis using System Tool Kit software (STK) [9] is performed to determine the accessibility and coverage of Morocco over time. The orbit parameters are presented in Table. I.

TABLE I
ORBIT PARAMETERS

Orbit parameter	Value
Orbit's Shape	Sun synchronous
Altitude	500Km
Inclination	97.79°
Right ascending node	185.78°

In this configuration, the satellite could perform 14 orbits per day with 03 accesses over the Moroccan maritime boundaries. Thus, the total coverage of this territorial sea can be achieved during these 03 passes as shown in Fig. 2. To maximize the capacity of collected AIS data, the needed passes over the required area will be reserved for AIS data acquisition, while only one access to the ground station implemented in Rabat will be used to download all the collected information, regarding the high data rate of S-band downlink. When the CubeSat starts AIS data housekeeping, the nadir pointing and vessel tracking with a VHF antenna are recommended for better reception. During no payload operation, the platform performs sun vector pointing to maximize power generation and aligns the S-band antenna with the ground station when visible.

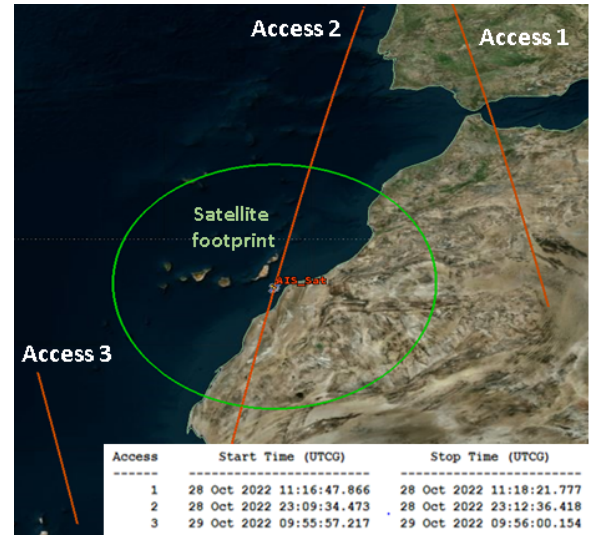


Fig. 2. Coverage analysis visualization.

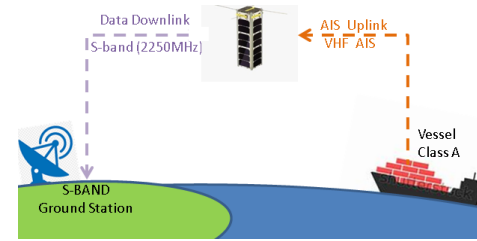


Fig. 3. CubeSat communication scheme.

III. AIS COMMUNICATION SYSTEM DESIGN

A. Spacecraft description and architecture

Fig. 3 shows the proposed CubeSat communication scheme, which allows AIS data reception using a Software Defined Radio and a S-band transceiver (2250MHz) using GMSK modulation for both subsystems. The proposed data transmission satellite's architecture includes the Electrical power subsystem (EPS), the Flight computer (FC), the payload subsystem and its controller (PC), and the communication subsystem. All of these subsystems are flight proven, connected via CAN interface and have an internal CSP based network. Fig. 4 illustrates the interaction between all subsystems.

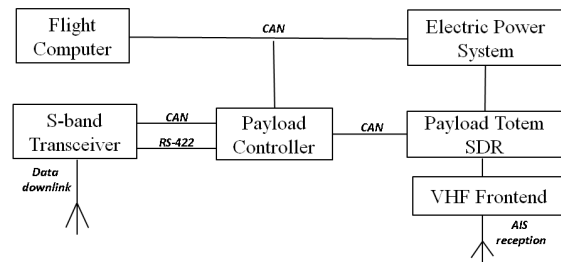


Fig. 4. Architecture diagram of Data transmission satellite.

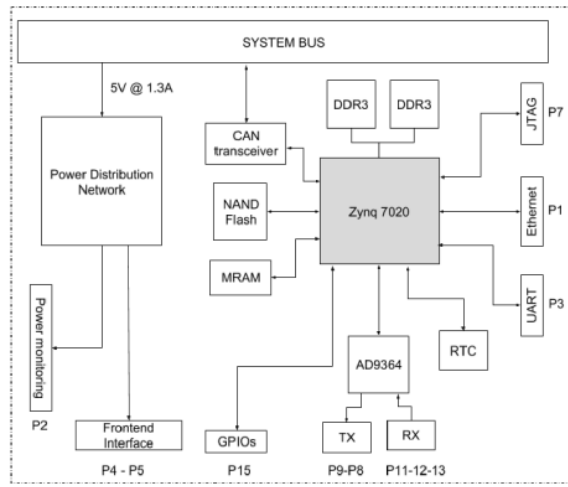


Fig. 5. Bloc diagram of TOTEM-Motherboard [7].

B. Payload TOTEM-Motherboard

The TOTEM-Motherboard is a Software Defined Radio developed for the nanosatellite's mission with respect to the standard of CubeSat. This hardware is based on a high-performance System-On-Chip (Xilinx Zynq-7000 series) and a wide-frequency range (70 MHz to 6GHz) RF transceiver (AD9364), as shown in Fig. 5. The VHF Frontend is interfaced with the TOTEM-Motherboard via an Rx connector as a simple integration solution using a piggyback board to save space. The reception chain is formed by a band-pass filter and an Low Noise Amplifier (LNA) in order to amplify the received signal.

The AIS service for data collection includes four parallel receivers for (AIS1, AIS2) and Long Range (AIS1, AIS2), configurable storage filters with optional packet deduplication, a garbage collector, and a data extraction dump filter. The data flow diagram needed to carry out AIS service using related telecommands is illustrated in Fig. 6. AIS messages are stored in the database using the selected filter after housekeeping collection. Messages extracted from the database can either be sent directly to the ground segment via space packets or saved to a file and downloaded later via the file channel.

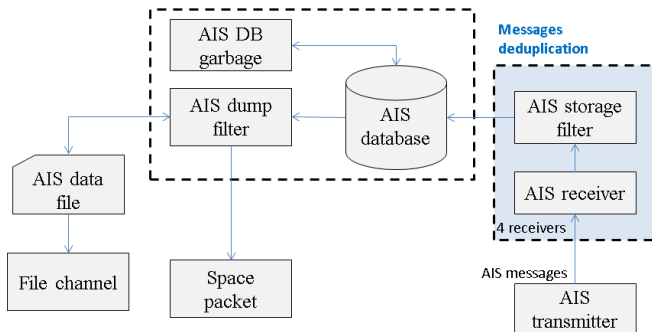


Fig. 6. AIS data flow diagram.

C. Data transmission subsystem

In this study, the data transmission subsystem relays on the selected S-band transceiver [11] matched with a Dual S-band Patch Antenna [10]. The Satlab SRS-3 is a full-duplex transceiver operating at S-band frequencies, with configurable output power, bit rate, and forward error correction (FEC). For modulation and link protocol, respectively, Minimum Shift keying (MSK) enhanced by a Gaussian filter and the CubeSat Space Protocol (CSP) are adopted. The antenna is designed to realize duplex satellite-to-ground links using a separated dual patch with circular polarization and a maximum gain of 6dBi.

D. Payload controller

The payload controller (PC) is in charge of establishing communication between the CubeSat and the payload by buffering data from the payload using a variety of interface connections, and transmitting the buffered data to the ground station via S-band transceiver.

IV. LINK BUDGET DESIGN

In satellite communication, the link budget is one of the main factors determining the reliability of transmission. This term describes a method for assessing the overall gains and losses related to the receiver, propagation channel, and transmitter. At the receiver side, the total power received P_{rx} expressed in dBW (1) is obtained by the Effective Isotropic Radiated Power, the gain of the receiver antenna G_{rx} , the free space path loss L_{fspl} and additional losses L_{tot} that include antenna pointing loss, polarization loss, cable and connector losses which are described in detail by [12].

$$P_{rx} = EIRP + G_{rx} - L_{tot} \quad (1)$$

here $EIRP = P_{tx} + G_{tx}$, P_{tx} and G_{tx} stand for the transmitter power and antenna gain, respectively. The free space path loss, which is a significant parameter in signal attenuation, is expressed by (2) in dB:

$$L_{fspl} = 20 \cdot \log_{10}(4\pi \cdot d \cdot \frac{f_c}{c}) \quad (2)$$

The distance between a satellite and a ground station or AIS terminal in Km is denoted by d , the speed of light in vacuum is denoted by c , and the carrier frequency in Hz is denoted by f_c . The Signal-to-Noise ratio (SNR) is the resulting power considering the effective noise temperature T_s of the receiver in Kelvin (3):

$$SNR = P_{Rx} - 10 \cdot \log_{10}(K \cdot B \cdot T_s) \quad (3)$$

where K is the Boltzmann Constant 1.38×10^{-23} J/K and B is the noise bandwidth (in hertz). In order to determine the link margin, a specific bit error rate (BER) that corresponds to an energy per bit of noise (E_b/N_0) required is desired depending on the modulation and Forward Error Correction (FEC).

$$Margin = SNR - required(E_b/N_0) \quad (4)$$

V. RESULTS AND DISCUSSIONS

In this section, the AMSAT-IARU Link Model Rev 2.5.5 spreadsheet [13] is used to design the proposed communication link. This model provides a complete tool to design a satellite communication link.

TABLE II
LINK BUDGET FOR UPLINK AIS CHANNEL 1

Parameter	Value	Unit
AIS transmitter		
Tx Power Output	12,5	watts
Tx Line Loss	1,0	dB
Tx EIRP	10,0	dBW
Propagation parameters		
Tx ant pointing Loss	0,3	dB
Polarization Loss	1,0	dB
Path Loss	143,1	dB
Atmospheric Losses	2,0	dB
Ionospheric Losses	0,5	dB
Signal Level at S/C	-136,9	dBW
Cubesat receiver		
Rx ant pointing Loss	0,7	dB
Rx antenna Gain	2,0	dBi
Rx line Losses	0,5	dB
Rx noise Temperature	390	K
SNR	66,9	dBHz
Data rate	9,6	Kbps
Eb/No	26,9	dB
Demodulation	GMSK	
BER	10^{-5}	
Eb/No threshold	10,6	dB
Margin	16,3	dB

TABLE III
LINK BUDGET FOR S-BAND DOWNLINK (2250MHz)

Parameter	Value	Unit
Cubesat transmitter		
Tx power output	1,0	watts
Tx line losses	0,5	dB
Tx antenna gain	6,0	dBi
Tx EIRP	5,5	dBW
Propagation parameters		
Tx ant pointing loss	0,5	dB
Polarization loss	0,5	dB
Path loss	165,0	dB
Atmospheric Loss	0,5	dB
Ionospheric loss	0,5	dB
Rain Loss	0,5	dB
Signal Level at GS	-162,0	dBW
Ground station		
Rx ant pointing loss	0,5	dB
Rx antenna gain	32,0	dBi
Rx line loss	0,5	dB
Rx Noise Temperature	500	K
SNR	71.1	dBHz
Data Rate	512	Kbps
Eb/No	14,1	dB
Demodulation	GMSK	
FEC	Cv(1/2,7) +RS(255,223)	
BER	10^{-5}	
Eb/No Threshold	7,6	dB
Margin	6,5	dB

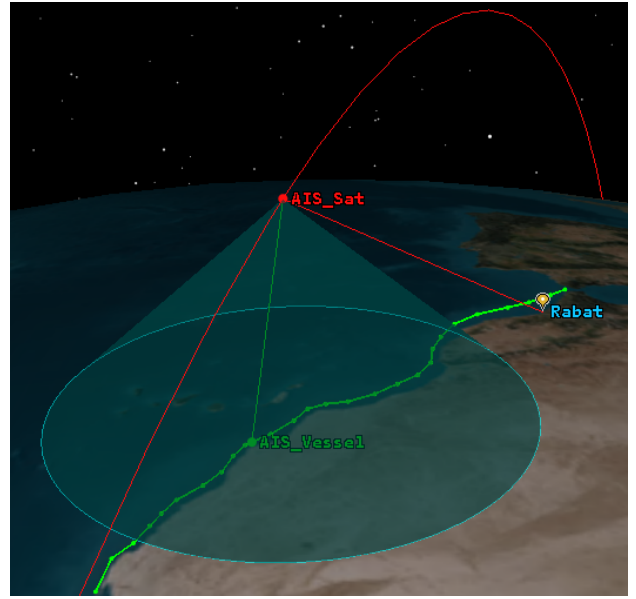


Fig. 7. 3D view of Vessel and Ground station access to the satellite

Tables. II and . III represent, respectively, at very low elevation angle (5° degree), the link budget results for AIS channel 1 uplink (161.975MHz) and S-band downlink (2250MHz). As known, Class A vessels radiate with 12.5W output power and transmit at a 9600bps datarate using GMSK modulation. Thus, the uplink signal level at the satellite is about -136.9dBW, which allows a sufficient link margin of 16.3dB to counter eventual losses.

At the ground station, the received signal is lower due to the limited power output of the CubeSat. Since the ground antenna gain reaches 32dB, the energy per bit to noise power density ratio E_b/N_o at the receiver is 14.1dB. Furthermore, the combination of GMSK modulation and concatenated code $R_{RS}R_{CC}$ lowers the E_b/N_o threshold to 7.6dB at the S-band transceiver. As a result, the downlink communication is assumed to satisfy the downlink requirement with a 6.5dB link margin.

After designing the communication link for the proposed concept of operation, a scenario of a CubeSat collecting AIS data from a vessel crossing the Moroccan territorial sea is simulated in STK as shown in Fig. 7.

The results obtained with STK software for the AIS channel 1 uplink and S-band downlink budgets are much closer to the calculated link budgets on the AMSAT IARU Link Model for a desired BER of 10^{-5} as presented in Table. IV.

TABLE IV
COMPARISON OF AMSAT AND STK LINK BUDGET

Parameter	Tool	FSPL(dB)	Margin(dB)
AIS 1 uplink	AMSAT	143.1	16.3
AIS 1 uplink	STK	143.4	16.0
S-band downlink	AMSAT	165.0	6.5
S-band downlink	STK	165.4	6.1

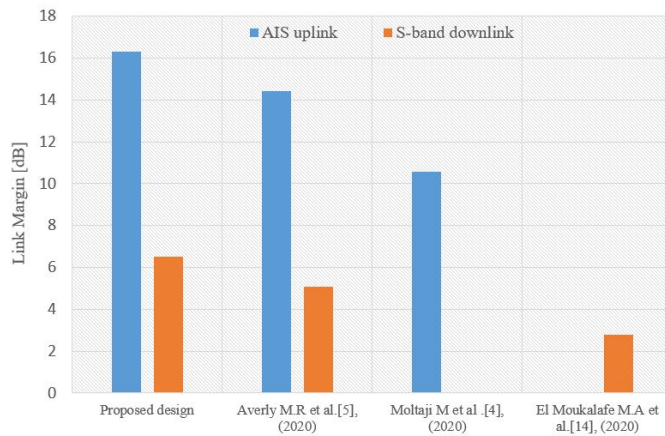


Fig. 8. Comparison between the proposed design and other works

To verify the obtained results, a comparison with other works on either AIS or S-band [14] communication link design was carried out. Fig. 8 shows better results of link margin for the proposed design compared with previous works. Therefore, the AIS link design presented in this paper is suitable for the mission requirements.

VI. CONCLUSION AND FUTURE WORK

In this paper, the preliminary design of a communication subsystem for a nanosatellite with a concept of operation for marine vessel monitoring over Morocco territorial sea has been done. The software defined radio payload subsystem and data transmission via S-band transceiver are selected for this mission. The satellite will be able to monitor vessels equipped by AIS terminals and transmit collected data to the ground station implemented in Rabat, Morocco. Regarding the communication subsystem, the link margin obtained by STK software and AMSAT Link Model reaches, respectively, 16.4dB and 6.4dB for AIS channel 1 uplink and S-band downlink. It is noticed that the link performances for the proposed design are adequate to mitigate various losses. As a future work, the proposed design can be adapted and implemented in orbit for experimental purposes.

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